

# Summary Report

## Titanic Dataset - Exploratory Data Analysis (EDA)

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### **OBJECTIVE:**

The goal of this project was to do an Exploratory Data Analysis on the dataset. For this purpose, I have chosen Titanic Dataset and performed EDA on it. This analysis helped me understand the data where I looked for patterns in the data and also studied that how different variables are related. My main aim was to find out which features most affect whether a passenger survives or not.

### **WORK PERFORMED:**

#### 1. Data Loading

- I imported the dataset using the Pandas library.
- The Titanic dataset was then displayed so I could understand what it looks like and what features are available.

#### 2. Missing Value Analysis

- I looked for missing values in each column of the dataset.
- I found missing values in the Age, Fare and Cabin columns.
- Then I filled in the missing values in Age, Fare using the right statistical methods after I looked at how the data is spread out.

#### 3. Data Preprocessing

- I changed columns like Age, Fare to the right type of numbers.
- The Titanic dataset was then ready for me to look at and do analysis.

#### 4. Exploratory Data Analysis

I made a lot of pictures to understand what the Titanic dataset is like including:

- Survival Count
- Survival by Gender
- Age Distribution
- Fare Distribution
- Fare Distribution by Passenger Class, which I showed as a boxplot

These pictures helped me see trends and compare groups of passengers.

#### 5. Correlation Analysis

- I picked the numbers from the dataset.
- Then I calculated the correlation matrix to see how the numbers are related to each other.
- I looked at how each feature's related to whether a passenger survived or not.

- I used the correlation values to rank the features by how much they affect whether a passenger survives.

## 6. Identification of Important Features

- From the correlation analysis I found the three features that have the strongest relationship with whether a passenger survives.
- I showed these features in a bar chart.

## 7. Scalability Analysis

- The project also talked about what would happen if the Titanic dataset got a lot bigger from 1,000 rows to 10 million rows.
- I thought about the problems that could happen like running out of memory taking long to compute pictures getting too complicated and needing more storage space.
- I suggested some solutions, like processing the data in chunks using computers that work together using storage formats that're efficient and taking samples of the data.

## **CONCLUSION:**

Analysis of the data was really helpful. It showed us problems with the data how the different features looked and how they relate to each other. We looked at which numerical features were most linked to people surviving.

This project shows how to do an analysis, including fixing the data drawing pictures of it doing statistical tests and checking features. The project also thinks about how to do this analysis, with datasets.

The exploratory data analysis was valuable because it helped us understand the data and make decisions. The Titanic dataset was analyzed to see what features were most associated with survival.